



Antibacterial activities of green synthesized gold nanoparticles

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ABSTRACT

Gold nanoparticles were synthesized using *Ananas comosus* as reducing agent. UV–visible spectra show the surface plasmon resonance peak at 546 nm. TEM measurement shows that the formation of little anisotropic nanoparticles with average size of 16 nm. Crystalline nature of the nanoparticles was evident from TEM images and peaks in the XRD pattern. The prepared gold nanoparticles reveal good antibacterial activity against gram positive and negative pathogens, which are found in water. So the prepared nanoparticles have found applications in water purification processes for inhibiting the growth of bacteria.

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1. Introduction

Metal nanoparticles have been received much attention due to their unique optical, electrical and catalytic properties. The size, shape and surface morphology of the particles were crucial in tuning these properties of nanosized metal particles. There are many synthesis methods have been developed to prepare metal nanoparticles, including chemical, physical and biological methods. Among these, biosynthesis of metal nanoparticles is the simplest and environment friendly method. Biosynthesis of nanoparticles using coriander, *Bischofia javanica* (L.), *Daucus carota*, *Solanum lycopersicum*, *Hibiscus cannabinus* leaf, *Moringa oliefera* flower, lemongrass, *Bacopa monnieri*, *Citrus unshiu* peel and *Ananas comosus* has been reported [1–10]. Among the different metallic nanoparticles, gold nanoparticles (AuNPs) exhibit novel properties. The chemical inertness and resistance to surface oxidation make gold an important material for use in nano-scale technologies and devices. This property is crucial when particle size approaches the nano-scale and the dominance of surface atoms results in an enhanced chemical reactivity.

Noble metal nanoparticles are used for the purification of water which is one of the essential enablers of life on earth. Water is one of the purest symbols of wealth, health, tranquility, beauty and originality. Waterborne pathogens including helminthes, protozoa, fungi, bacteria, rickettsiae, viruses and prions, can cause many diseases. In India 80% of the diseases are due to bacterial contamination of drinking water. To protect the water purity, the removal or deactivation of pathogenic bacteria in water is very important. Antibacterial activities of nanoparticles against various

pathogens have also been established [11–14]. In the present study, the antibacterial assay was done on various pathogenic bacteria like *Staphylococcus aureus* and *Pseudomonas aeruginosa*, which are commonly found in water. In this work, AuNPs were synthesized using *A. comosus* fruit extract as reducing agent. Since *A. comosus* is a commonly available fruit and good source of citric acid, malic acid and ascorbic acid [15]. Here the size and aggregation of the nanoparticles were controlled without additional stabilizing agents. The antibacterial activity of prepared AuNPs has been also described.

2. Experimental techniques

For the synthesis of AuNPs, 5 ml of *A. comosus* extract was added to aqueous solution of HAuCl_4 (6 mM) and stirred continuously for 5 min and it was turned to dark ruby red after 6 h which gives AuNPs (g1). This colloid was found to be stable for about one month showing no precipitation or color change. The absorption spectrum of the g1 was measured using a Shimadzu spectrophotometer (UV 1700) in 300–800 nm range. After one month the solution was dried at 100 °C. The dried powder (g1) was taken for further characterizations such as X-ray diffraction (XRD), Transmission Electron Microscope (TEM) and Energy Dispersive X-ray Spectroscopy (EDX). XRD analysis of the g1 was done using PANalytical X'pert – PRO diffractometer with $\text{Cu K}\alpha$ radiation. TEM analysis was done using a JEOL JEM 2100 High Resolution TEM. The antibacterial assay was done on various pathogenic bacteria like *S. aureus* and *P. aeruginosa*. Nutrient agar medium was used to cultivate bacteria. The dried powder of g1 was taken at the concentration of 50 $\mu\text{g/ml}$ for the antibacterial tests. The dried powder of *A. comosus* extract (50 $\mu\text{g/ml}$) was used as control. These plates were incubated at 37 °C for 24 h. Then inhibitory

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action of AuNPs on the growth of the bacteria was determined by the measurement of diameter of inhibition zone.

3. Results and discussion

The stability of the synthesized AuNPs was studied by measuring its optical spectra of g1 at different time, since it was an important parameter in synthesis of nanoparticles. Fig. 1(i) shows the UV–vis spectra of g1 at different reaction time. The visual inspection of g1 at different reaction time has been also shown in Fig. 1(i) inset. As the effect of reaction time, the surface plasmon resonance (SPR) bands of the prepared AuNPs exhibited blue shift (550–546 nm) with increasing absorbance. The blue shifted and narrow SPR band indicating the formation of small spherical nanoparticles with the effect of reaction time was also found in TEM images. The absorbance value of the band increases with increasing time period, which indicates that the continued reduction of metal ions and concentration of nanoparticles increased. The formation of AuNPs started within 10 min, and then the intensity of the SPR peak increased up to 6 h, after that there was no change in SPR peak for 30 days. After 6 h, the availability of a larger number of nuclei induces a decrease in the particle size, because smaller metal nuclei grow and consume metal ions at the same time. The stable positions of the SPR and color of the colloids remain stable for 30 days without any sign of aggregation of the particles. This stability results from a potential barrier that develops as a result of the competition between weak Van der Waals forces of attraction and electrostatic repulsion.

Fig. 1(ii) shows the XRD pattern for the dried powder of *A. comosus*. Three diffraction peaks were observed at 28.5°, 40.8° and 50.9° signify the presence of ascorbic acid (JCPDS 22-1560), citric acid (JCPDS 22-1568) and malic acid (JCPDS 23-1631) in the *A. comosus* extract. Fig. 1(ii)(b) shows the XRD pattern for g1. The diffraction peaks were observed at 38.2°, 44.1°, 64.8° and 77.6° in the 2θ range and they corresponding to (111), (200), (220) and (311) Bragg's reflections based on the FCC structure of AuNPs with space group of Fm-3m (JCPDS: 04-0784). No peaks of crystallographic impurities in the sample have been found. The obtained broader peaks are directly proportional to the smaller particle size. The average particle size (D), lattice constant (a), cell volume (V), surface area to volume (SA:V) ratio, specific surface area (SSA) and crystallinity index (I_{cry}) were calculated and tabulated Table 1). The calculated value of crystallinity index of g1 shows monocrystalline nature.

TEM image of g1 was shown in Fig. 2(a). It shows the formation of smaller nanoparticles of spherical, decahedron and little triangles in the range of 5–20 nm with an average of ~16 nm. Decahedron (Fig. 2(a) (inset)) is a strained structure but possesses a slightly more efficient space-filling structure. It is an equilibrium shape bound completely by triangular (111) facets and can be thought of as five tetrahedral sharing a common edge along a fivefold axis. The presence of multiply twinning in AuNPs observed in Fig. 2(a) (inset), which exhibit near fivefold symmetry. FCC structured nanocrystalline materials have a tendency to nucleate and grow into twinned particles with their surfaces bounded by lowest energy facets (111) [16]. The nanotriangles formed as a result of rapid reduction, assembly and room temperature sintering of spherical nanoparticles, rearrangement and aggregation of smaller size particles. The formation of the gold nanotriangles act as a nuclei for further growth into anisotropic triangular structures. The excess free energy per unit area for a particular crystallographic face is surface energy which determines the faceting and crystal growth for nanoparticles. The nanotriangles experienced a shrinking process to reduce their high surface energy to a minimum resulting in the formation of blunt-angled nanotriangles from sharp-angled nanotriangles. Fig. 2(b) shows the EDX analysis of g1. It confirms the formation of gold atoms in the AuNPs and reveals higher counts at 2.2 keV due to gold nanocrystallite absorption due to SPR metallic gold nanocrystals.

Nanoparticles applications in medicine depend on the size and the composition of the nanoparticles. The ability to target diverse bacterial structures was the important property of nanoparticles. AuNPs have a great bactericidal effect and it possess well-developed surface chemistry, chemical stability and appropriate smaller size which make them easier to interact with the micro-organisms. The antibacterial activity of g1 against gram positive (*S. aureus*) and gram negative pathogen (*P. aeruginosa*) was shown in Fig. 3(a)–(c).

Based on Fig. 3, it can be concluded that *P. aeruginosa* was more active and inhibits greatest bacterial effect. The differential

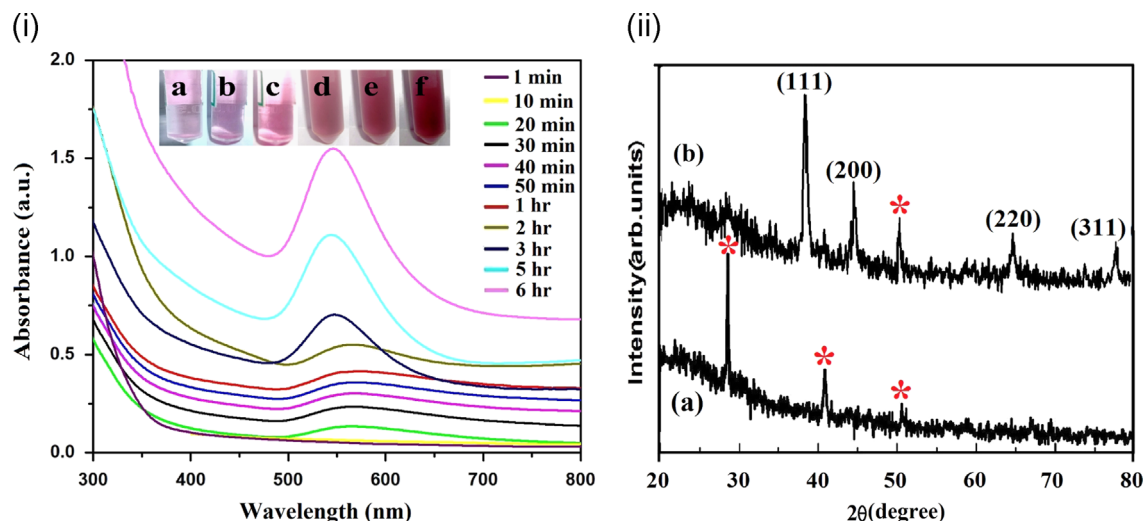


Fig. 1. (i) Optical absorption spectra and color changes (inset) of g1 at different reaction time in reaction system (a, b, c, d, e and f vs 10 min, 20 min, 40 min, 1 h, 3 h and 6 h) and (ii) XRD pattern for *A. comosus* (a) and g1 (b) (* due to *A. comosus* fruit extract).

Table 1

The average particle size, lattice constant, cell volume, SA: V ratio, SSA and I_{cry} values of g1.

Prepared AuNPs	D (nm)	a (Å)	V (Å ³)	SSA (m ² /g)	SA:V ratio	I_{cry}
g1	18.32	4.0720	67.52	16.96	0.32	~0.8733

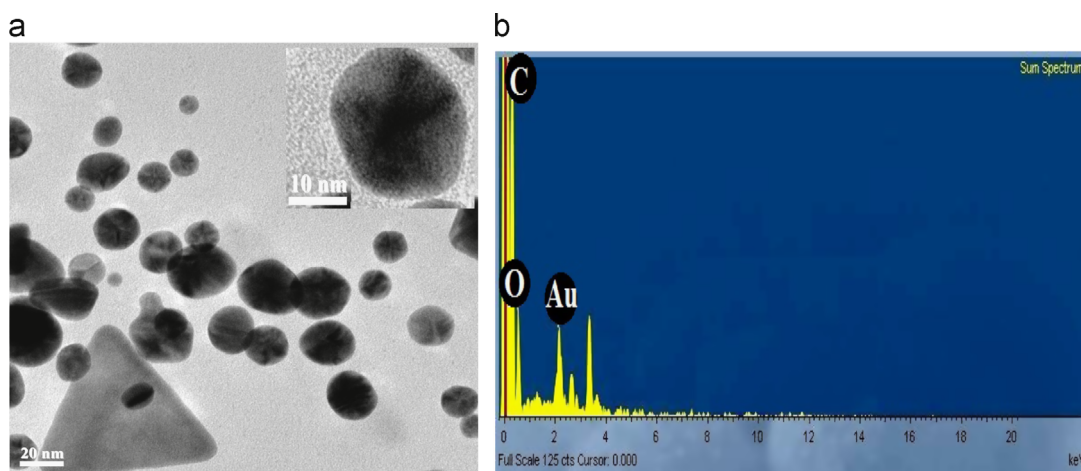


Fig. 2. TEM micrograph (a) and EDX graph (b) of g1.

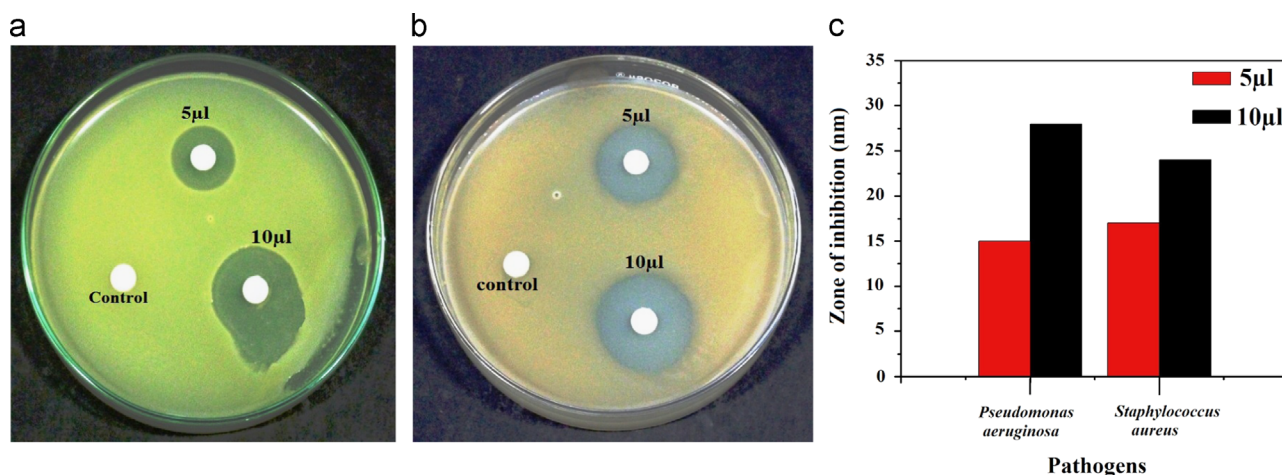


Fig. 3. Zone of inhibition of g1 against *Pseudomonas aeruginosa* (a) *Staphylococcus aureus* (b), and antibacterial activity of g1 (c).

sensitivity of gram negative and gram positive bacteria towards AuNPs possibly depends upon their cell structure, physiology, metabolism and their interaction with the charged AuNPs. The antibacterial effect of AuNPs on microorganisms may be held through the electrostatic attraction of positive charged gold and negative charged cell surface of microorganism. Small AuNPs are able to be penetrated inside the bacteria and caused further damage, lose their activity and finally cell death. At last, the AuNPs release gold ions, which will have an additional contribution to the antibacterial activity of the AuNPs. Smaller particles having larger SSA can increase the ability to penetrate cell membrane and give more bactericidal effect. Because SSA has a particular importance in reactivity. It gives the rate at which the reaction will proceed. The large number of atoms available in g1 makes the reaction faster and hence makes them more suitable for broad kind of applications. This characteristic enhances biological and chemical activity of the nanoparticles with high antibacterial efficacy.

4. Conclusion

The present simple study was designed to slow reduction of chloroauric acid using fruit extract of *A. comosus* as reducing agent. The SPR of prepared AuNPs was confirmed by UV–visible spectral

analysis. As the reaction time increases, absorption spectra shows blue shift with decreasing particle size. The prepared AuNPs reveal good antibacterial activity against gram positive and negative pathogens, which are found in water. So the prepared nanoparticles have found applications in water purification processes for inhibiting the growth of bacteria.

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